Modeling Relationship between the PMT Response and the Characteristics of Air Shower Particles

for SWGO Using Score-Based Diffusion Model

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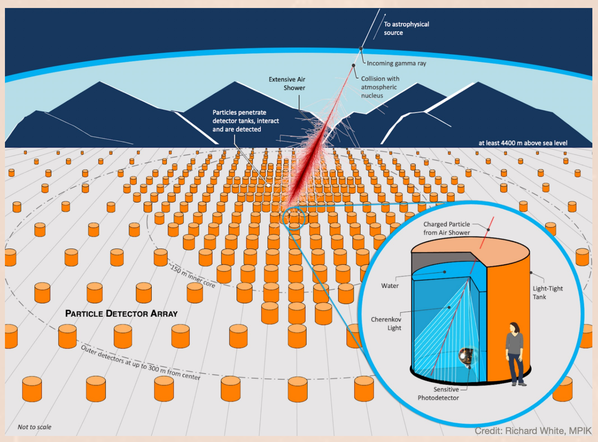
SWGO

The Southern Wide field Gamma-ray Observatory (SWGO) will be the **first ground-based gamma-ray sampling array located in the southern hemisphere**, providing a high duty cycle and wide field of view. To be located in the Atacama Astronomical Park, Chile, at an altitude of 4,770 meters, SWGO is designed to detect gamma rays in the energy range of 100 GeV to 1 PeV by detecting Cherenkov light produced by extensive air shower particles in water tanks. With this technique, SWGO aims to explore **very-high-energy (VHE) to ultra-high-energy (UHE) gamma-ray sources**, including those that have never been observed before.



Motivation

Simulation in SWGO consists of two parts:

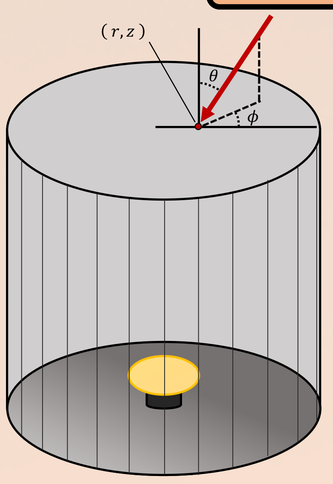
1. **CORSIKA** : models the extensive air shower produced by the primary particle.
2. **HAWCSim** : bases on GEANT4, simulating Cherenkov photons generated through the interaction between a secondary particle and an individual detector, ultimately producing the corresponding PMT response.

By aggregating the PMT responses from each detector from the entire array, the complete array-level simulation result is obtained.

However, at ultra-high energies, the number of secondary particles increases exponentially, making full detector simulation with GEANT4 **computationally too expensive.**

To address this limitation, we propose a **Score-Based Conditional Diffusion Model** that learns direct mapping from the secondary particle's characteristics to PMT responses, thereby bypassing repeated Cherenkov light simulations.

Score-Based Diffusion Model



Secondary particle's characteristics (c) :

$$c = [\,\mathrm{ID}, E, \theta, \phi, r, z]$$

ID : particle type (muon, gamma, electron)
E : energy
θ, ϕ : incoming zenith, azimuth angle
r, z : radius, height of the intersection

The PMT response (x_0) :

$$x_0 = [\,\mathrm{NPE}, \text{First Hit Time}]$$

NPE : Total number of detected photoelectrons
First Hit Time : Arrival time of the earliest detected photoelectron

Main Objective :

Train conditional score-based diffusion model to learn the posterior:

$$p(x \mid c)$$

Forward Process :

The PMT response x_0 is gradually perturbed using a noise schedule $\{\beta_t\}_{t=1}^{T}$, forming

$$q(x_t|x_0) = \mathcal{N}\left(x_t \mid \sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t I)\right), \quad \overline{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$$

Training Purpose :

A neural network is trained to predict added noise:

$$\mathcal{L}_{\mathrm{DSM}} = \mathbb{E}_{x_0,\epsilon,t}\left[\,\|\epsilon - \epsilon_\theta(x_t, t, c)\|^2\,\right]$$

Score Function:

From Tweedie – theorem, we can approximate the score using trained network

$$s_\theta(x_t, t, c) = -\frac{1}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t, t, c) \approx \nabla_{x_t}\log p(x_t \mid c)$$

Reference

[1] Heck et al., "CORSIKA: A Monte Carlo Code to Simulate Extensive Air Showers," 1998.\par
[2] Agostinelli et al., "Geant4: A Simulation Toolkit," Nucl. Instrum. Methods A, 2003.\par
[3] Huentemeyer et al., "The Southern Wide-Field Gamma-Ray Observatory (SWGO)," Nucl. Instrum. Methods A, 2023.
[4] Ho et al., "Denoising Diffusion Probabilistic Models," NeurIPS, 2020.

Sampling with Classifier-Free Guidance

Conditional Sampling:

To generate a sample from the posterior $p(x_0 \mid c)$, we start with some random Gaussian noise $x_T \sim \mathcal{N}(0, I)$, and iteratively denoise it using Langevin dynamics

$$x_{t-1} = x_t + \frac{\beta_t}{2}s_\theta(x_t, t, c) + \sqrt{\beta_t}z, \qquad z \sim \mathcal{N}(0, I)$$

the reverse process proceeds from $t = 1000 = T$ to $t = 1$

Classifier-Free Guidance (CFG):

To improve conditional generation quality, Classifier-Free Guidance score

$$s_{CFG} = s_\theta(x_t, t, c) + w \cdot \left(s_\theta(x_t, t, c) - s_\theta(x_t, t, \emptyset)\right), \quad w = 2$$

Using the CFG score, the update becomes :

$$x_{t-1} = x_t + \frac{\beta_t}{2}s_{CFG}(x_t, t, c) + \sqrt{\beta_t}z, \qquad z \sim \mathcal{N}(0, I)$$

Sampling Algorithm:

Initialize $x_T \sim \mathcal{N}(0, I)$, $T = 1000$
for $t = 1000, 999, \ldots, 1$:
  Compute CFG score $s_{CFG}(x_t, t, c)$
  Update: $x_{t-1} = x_t + \frac{\beta_t}{2}s_{CFG}(x_t, t, c) + \sqrt{\beta_t}z, \qquad z \sim \mathcal{N}(0, I)$
Return x_0

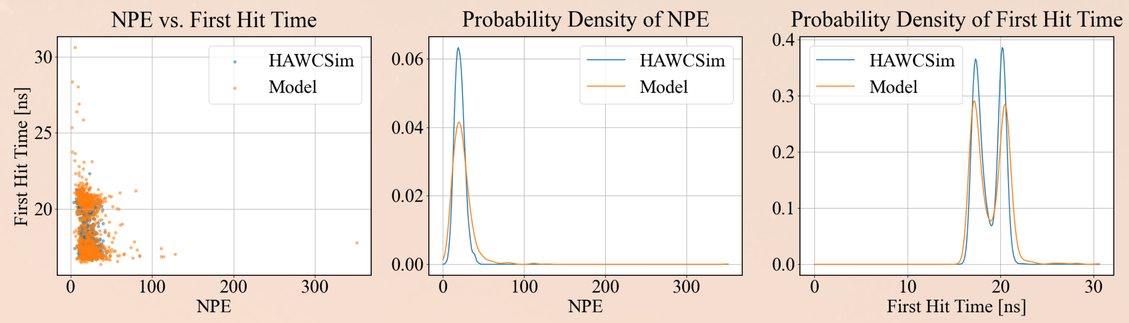
Final Output :

$x_0 = [\,\mathrm{NPE}, \text{First Hit Time}]$ consistent with the given particle condition c

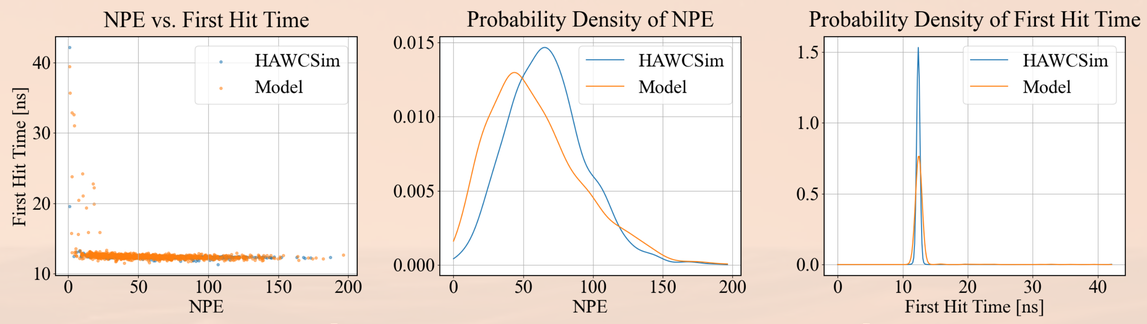
Model Performance

The following compares 1000 samples generated by the model conditioned on a given c with the actual 1000 results from the actual HAWCSim simulation.

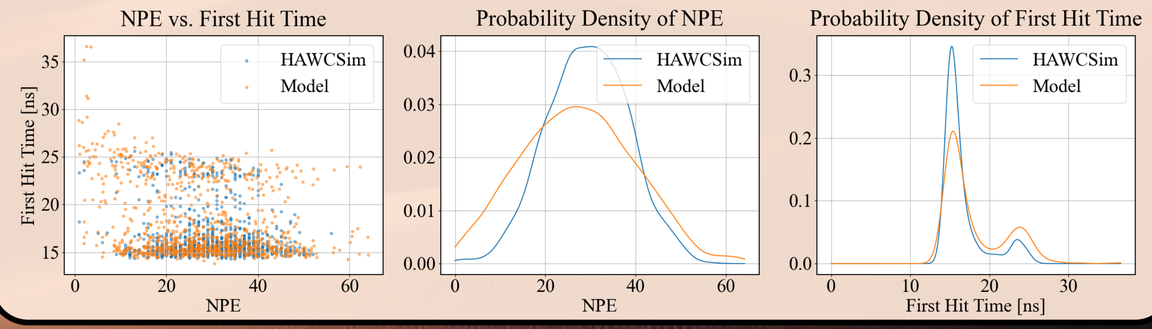
$c = [\,\mathrm{muon}, 3\mathrm{GeV}, 30°, 80°, 208\mathrm{cm}, 412.4\mathrm{cm}]$



$c = [\,\mathrm{electron}, 1\mathrm{GeV}, 30°, 65°, 260\mathrm{cm}, 217.053\mathrm{cm}]$



$c = [\,\mathrm{gamma}, 1\mathrm{GeV}, 55°, 80°, 78\mathrm{cm}, 412.4\mathrm{cm}]$



Conclusion & Outlook

- The model **successfully captures the relationship** between the **PMT response** and the **secondary particle's characteristics** across a wide range of conditions.
- Some discrepancies remain, which will be addressed through future fine-tuning.

Acknowledgements

National Research Foundation of Korea (RS-2023-00280210)